TIANMING HUO

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Education

National University of Singapore Aug 2020 – Jan 2025
Ph.D. in Industrial System Engineering and Management
with NUS Research Scholarship
Advisor: Cheung Wang Chi
Research Area: Operations Research, Revenue Management, Online Algorithms, Competitive Analysis, Resource Allocation, Order Scheduling, Dynamic Pricing

National University of Singapore Aug 2015 – June 2019
B.Eng. in Industrial System Engineering and Management
Graduated with Honours (Distinction)

Research Papers

Optimal Online Allocation of One-Way Substitutable Resources 2026
Tianming Huo, Guodong Lyu, Wang Chi Cheung
under review at *Operations Research*
Available on SSRN
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=7109699

Online Reusable Resource Allocations with Multi-class Arrivals 2022
Tianming Huo, Wang Chi Cheung
Forthcoming in *Operations Research*
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4320423

Online Lead Time Quotation with Predictions 2024
Tianming Huo, Wang Chi Cheung
forthcoming in *INFORMS Journal of Computing*
https://pubsonline.informs.org/doi/10.1287/ijoc.2025.1104

Optimal Markdown Pricing with Patient Customers 2025
Wang Chi Cheung, Tianming Huo
Major revision (2st round) for Production and Operations Management (POM)
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5522978

Online Reusable Resource Assortment Planning with Customer-Dependent Usage Durations
Tianming Huo, Wang Chi Cheung 2023
Available on SSRN
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4504713

Principal Component Analysis for High Dimension Stochastic Gaussian Process Model Fitting
Maxime Xuereb, Tianming Huo and Szu Hui Ng 2019

2019 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)
https://ieeexplore.ieee.org/abstract/document/8978547

Invited Presentations

Online Reusable Resource Allocations with Multi-class Arrivals

- 2022 INFORMS Revenue Management and Pricing (RMP) Section Conference June, 2022
  Virtual
- 2022 Analytics for X Conference Sept, 2022
  Singapore
- 2023 POMS-HK International Conference Jan, 2023
  Virtual
- 2025 INFORMS International Conference (Advance booking) Jul, 2025
  Singapore
- 2026 POMS-HK International Conference (Advance booking) Jan, 2026
  Hong Kong
- 2026 INFORMS Annual Meeting Nov, 2026
  San Francisco, United States

Online Reusable Resource Assortment Planning with Customer-Dependent Usage Durations

- 2023 INFORMS Manufacturing and Service Operations Management (MSOM) Conference
  Montreal, Canada June, 2023
- 2023 INFORMS Revenue Management and Pricing (RMP) Section Conference July, 2023
  London, United Kingdom
- 2023 INFORMS Annual Meeting Oct, 2023
  Phoenix, Arizona, United States
- 2024 POMS-HK International Conference Jan, 2024
  Hong Kong

Online Lead Time Quotation with Predictions

- 2024 INFORMS Revenue Management and Pricing (RMP) Section Conference July, 2024
  Los Angeles, United States

Optimal Markdown Pricing with Patient Customers

- 2024 INFORMS Revenue Management and Pricing (RMP) Section Conference July, 2024
  Los Angeles, United States

Optimal Online Allocation of One-Way Substitutable Resources

- 2026 INFORMS Annual Meeting Nov, 2026
  San Francisco, United States

Teaching

Teaching Assistant, National University of Singapore

- IE2100: Probability Models with Applications AY20/21, AY21/22, AY22/23 Sem 2
  Teaching tutorial sessions, Grading homework, Holding office hours
- IE3101: Statistics for Engineering Applications AY21/22 Sem 1
  Teaching tutorial sessions, Grading homework, Holding office hours
- IE5123: Reliability Engineering AY22/23 Sem 1
  Grading Homework, Holding Office hours
- IE5121: Quality Planning and Management AY23/24 Sem 1
  Teaching tutorial sessions, Grading Homework, Holding Office hours

Reviewing Service

Reviewer for *Management Science, Transportation Science*

Working Experience

Research Fellow, NRF Postdoctoral Award Recipient May 2026 – Present
Singapore Management University

- Awarded the Singapore National Research Foundation (NRF) Postdoctoral Award, the only recipient in the College of Integrative Studies: four years of funding (to March 2030) with a S$250,000 research grant for independent research. Lead research projects for the topic of Algorithmic Resilience in Smart City Operations: Real-time Resource Allocations for Urban Sustainability. Design real-time algorithms that incorporate ML-learned predictions to enable intelligent decision-making that is both data-informed and resilient against sudden disruptions.

Research Fellow
National University of Singapore April 2025 – May 2026

- Lead a research project to optimize the real-time allocation of resources under realistic settings, including designing algorithms, deriving theoretical results and conducting numerical experiments.

Capacity Industrial Engineer Aug 2019 – July 2020
Micron Semiconductor Asia Pte Ltd

- Achieved history high weekly backend throughput by improving workstation production rate and ensuring sufficient capacity at the Encapsulation/Post-Encapsulation segment of the line
- Identified capacity bottleneck and procured new workstation to improve overall capacity
- Coordinated workstation relocation project while ensuring no production impact
- Trained new team member on capacity IE calculation and model operation
- Rectified, updated and maintained capacity model to reflect precise data on production line

System Design Project Team Member Aug 2018 – Apr 2019
Softbank Robotics Singapore

- Increased footfall to shops experiencing low volume of customers through the design and deployment of a humanoid robot programme

Central Procurement Intern Jan 2018 – Jun 2018
Rolls-Royce Singapore Pte Ltd

- Developed Excel VBA macro tools to automate and optimize data analysis and reporting

Skills

Software Skills: Proficient in Python, Matlab. Familiar with C/C++, R
Language: Chinese (Native), English (Fluent)